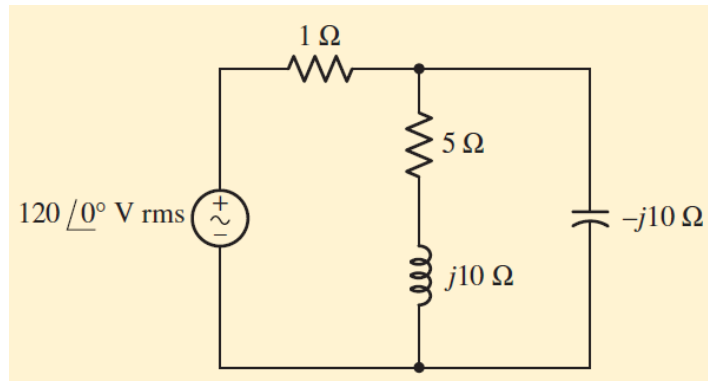
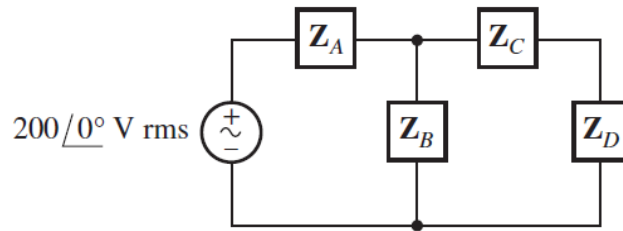




Set A

Find the complex power absorbed by the (a) $1\ \Omega$ resistor, (b) $-j10\ \Omega$ capacitor, and (c) source. (d) Find the instantaneous power provided by the source in Watts.



Set B

Given $Z_A = 5 - j2\ \Omega$, $Z_B = 3\ \Omega$, $Z_C = 8 + j4\ \Omega$, and $Z_D = 15 - j30\ \Omega$.

- Find the average power delivered to each load.
- Find the power factor at which the source operates.